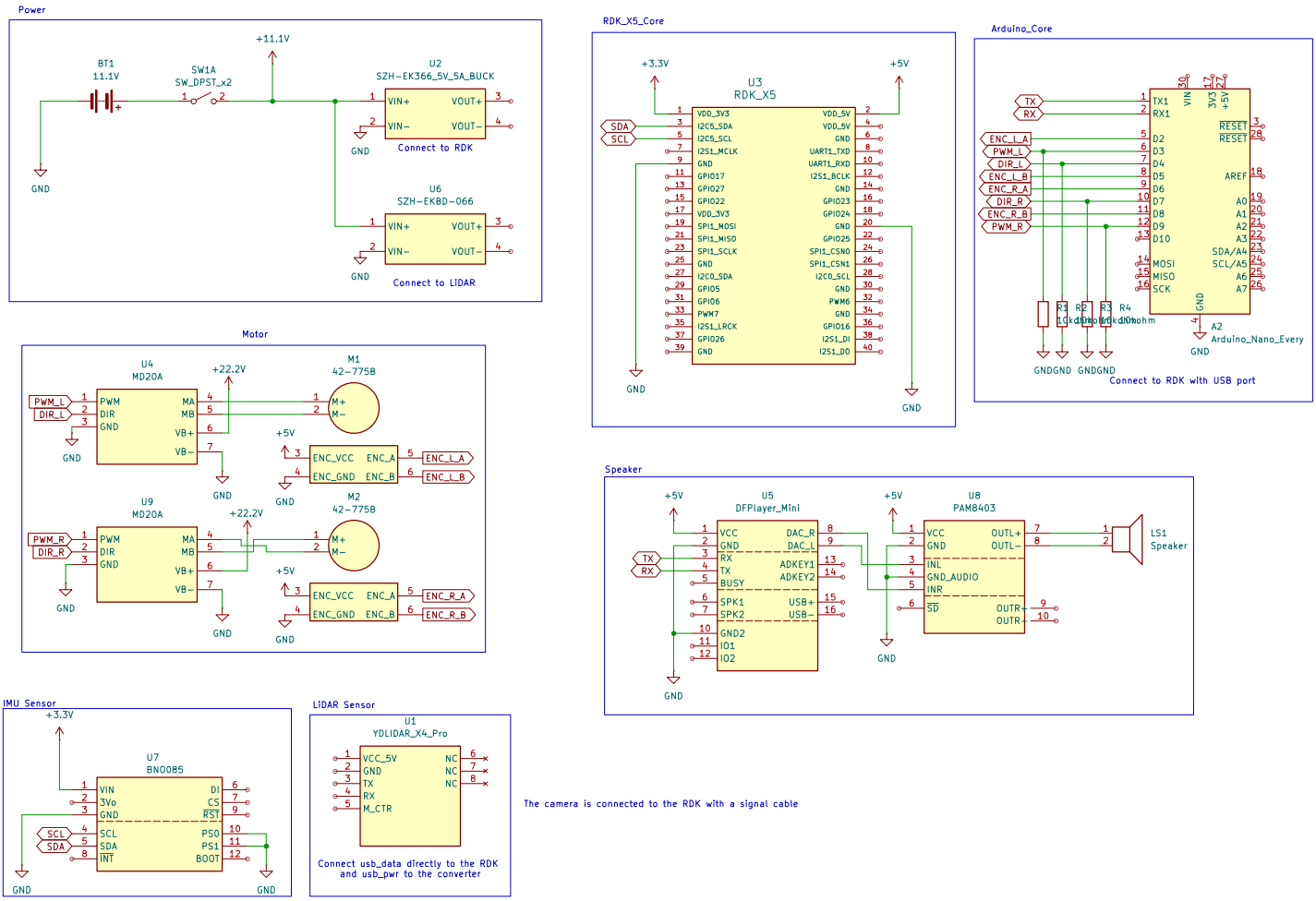

Power
BT1
11.1V
SW1A
SW_DPST_x2
+11.1V
U2
SZH-EK366_5V_5A_BUCK
VIN+
VIN-
VOUT+
VOUT-
GND
Connect to RDK
U6
SZH-EKBD-066
Connect to LiDAR
RDK_X5_Core
U3
RDK_X5
+3.3V
+5V
SDA
SCL
Arduino_Core
TX
RX
ENC_L_A
PWM_L
DIR_L
ENC_L_B
ENC_R_A
DIR_R
ENC_R_B
PWM_R
R1
R2
R3
R4
10kohm
A2
Arduino_Nano_Every
Connect to RDK with USB port
Motor
U4
MD20A
+22.2V
M1
42-775B
PWM_L
DIR_L
ENC_VCC
ENC_GND
ENC_A
ENC_B
ENC_L_A
ENC_L_B
U9
MD20A
M2
42-775B
PWM_R
DIR_R
ENC_R_A
ENC_R_B
Speaker
U5
DFPlayer_Mini
U8
PAM8403
LS1
Speaker
IMU Sensor
U7
BNO085
LiDAR Sensor
U1
YDLIDAR_X4_Pro
Connect usb_data directly to the RDK
and usb_pwr to the converter
The camera is connected to the RDK with a signal cable